

Problem

For an ideal op amp, which of the following statements are not true?

- (a) The differential voltage across the input terminals is zero.
- (b) The current into the input terminals is zero.
- (c) The current from the output terminal is zero.
- (d) The input resistance is zero.
- (e) The output resistance is zero.

Step-by-step solution

Step 1 of 3

The typical ranges of op-amp parameters are:

The ideal open loop gain of the op-amp is infinite.

The input resistance of the op-amp is ideally infinite.

The output resistance of the op-amp is zero.

Step 2 of 3

The two important characteristics of the ideal op-amp are:

1. Due to infinite input resistance between the input terminals, an open circuit exists implies, the current cannot enter the op-amp. The currents into both input terminals are zero.

2. The voltage across the input terminals is equal to zero.

Thus, the differential voltage across the input terminals is zero.

Step 3 of 3

Therefore, by observing parameters and characteristics of an ideal op amp, conclude that option (c) and option (d) were wrong.

Therefore, the statements (c) and (d) are not true.